The valence commutator of the CT generator

$[\hat{V}, \hat{A}]$, the $(2,1)$ and $(2,2)$ blocks

This continues `ct_F_A.tex` for the valence part of the Hamiltonian, in the same convention: the crossed Fock order, excitation operators, and general indices ranging over the whole space $\{i\}\cup\{a\}$,

$$\hat{V} = \tfrac{1}{2}\, g^{pq}_{rs}\, \hat{E}^{rs}_{pq}, \qquad \hat{T}_1 = t^a_i\, \hat{E}^a_i, \qquad \hat{T}_2 = \tfrac{1}{2}\, t^{a\beta}_{ij}\, \hat{E}^{a\beta}_{ij}, \tag{1}$$

with g^{pq}_{rs} antisymmetric in (p,q) and in (r,s), and $t^{a\beta}_{ij}$ antisymmetric in (a,β) and in (i,j). As for $\hat{F}$, the tensor of $\hat{V}$ carries its upper pair onto the operator's lower pair.

1. The reduction

$$[\hat{V}, \hat{A}] = [\hat{V}, \hat{T}] + [\hat{V}, \hat{T}]^\dagger = [\hat{V}, \hat{T}_1] + [\hat{V}, \hat{T}_2] + \text{h.c.}, \tag{2}$$

so the valence transformation needs two blocks: the $(2,1)$ block $[\hat{V}, \hat{T}_1]$ and the $(2,2)$ block $[\hat{V}, \hat{T}_2]$. Both are computed below and both are kept with all their terms, as in the Fock case: each term is a different operator with its own index pattern.

2. The $(2,1)$ block $[\hat{V}, \hat{T}_1]$

$$[\hat{V}, \hat{T}_1] = \tfrac{1}{2}\, g^{pq}_{rs}\, t^a_i\, [\hat{E}^{rs}_{pq}, \hat{E}^a_i], \tag{3}$$

and the $(2,1)$ commutator is minus the $(1,2)$ one, which is the rule already verified in `ct_F_A.tex`:

$$[\hat{E}^q_p, \hat{E}^{tu}_{rs}] = \delta^t_p\, \hat{E}^{qu}_{rs} + \delta^u_p\, \hat{E}^{qt}_{sr} - \delta^q_r\, \hat{E}^{tu}_{ps} - \delta^q_s\, \hat{E}^{ut}_{pr}. \tag{4}$$

Interchanging the two operators reverses the sign, and applying (4) with $\hat{E}^a_i$ in the first slot,

$$\begin{aligned} [\hat{E}^{rs}_{pq}, \hat{E}^a_i] &= -[\hat{E}^a_i, \hat{E}^{rs}_{pq}] \\ &= -\delta^r_i\, \hat{E}^{as}_{pq} - \delta^s_i\, \hat{E}^{ar}_{qp} + \delta^a_p\, \hat{E}^{rs}_{iq} + \delta^a_q\, \hat{E}^{sr}_{ip}. \end{aligned} \tag{5}$$

Each delta contracts one index of g, and the block keeps its four terms:

$$\boxed{\begin{aligned} [\hat{V}, \hat{T}_1] = \tfrac{1}{2}\, g^{pq}_{rs}\, t^a_i \Big(&+ \delta^a_p\, \hat{E}^{rs}_{iq} + \delta^a_q\, \hat{E}^{sr}_{ip} \\ &- \delta^r_i\, \hat{E}^{as}_{pq} - \delta^s_i\, \hat{E}^{ar}_{qp} \Big). \end{aligned}} \tag{6}$$

The first two terms carry the virtual index of the amplitude on the operator's upper pair, the last two the occupied index on the lower pair. Contracting the deltas with g gives the coefficient form,

$$[\hat{V}, \hat{T}_1] = \tfrac{1}{2}\, t^a_i \Big(g^{aq}_{rs}\, \hat{E}^{rs}_{iq} + g^{pa}_{rs}\, \hat{E}^{sr}_{ip} - g^{pq}_{is}\, \hat{E}^{as}_{pq} - g^{pq}_{ri}\, \hat{E}^{ar}_{qp} \Big), \tag{7}$$

again with no cancellation between the terms.

3. The $(2,2)$ block $[\hat{V},\hat{T}_2]$

$$[\hat{V},\hat{T}_2] = \tfrac{1}{4}\, g^{pq}_{rs}\, t^{a\beta}_{ij}\, [\hat{E}^{rs}_{pq}, \hat{E}^{a\beta}_{ij}], \tag{8}$$

the $\frac{1}{4}$ being the two halves, one from $\hat{V}$ and one from $\hat{T}_2$. The $(2,2)$ rule in the paired order is

$$\begin{aligned}
[\hat{E}^{rs}_{pq}, \hat{E}^{uw}_{tv}] = {} & + \delta^u_p\, \hat{E}^{rsw}_{tqv} + \delta^w_p\, \hat{E}^{rsu}_{vqt} + \delta^u_q\, \hat{E}^{srw}_{tpv} + \delta^w_q\, \hat{E}^{sru}_{vpt} \\
& + \delta^u_p \delta^w_q\, \hat{E}^{rs}_{tv} + \delta^w_p \delta^u_q\, \hat{E}^{rs}_{vt} \\
& - \delta^r_t\, \hat{E}^{uws}_{pvq} - \delta^s_t\, \hat{E}^{uwr}_{qvp} - \delta^r_v\, \hat{E}^{wus}_{ptq} - \delta^s_v\, \hat{E}^{wur}_{qtp} \\
& - \delta^r_t \delta^s_v\, \hat{E}^{uw}_{pq} - \delta^s_t \delta^r_v\, \hat{E}^{uw}_{qp}.
\end{aligned} \tag{9}$$

Applying it with $(a\beta ij)$ in the second slot, $u=a$, $w=\beta$, $t=i$, $v=j$,

$$\boxed{\begin{aligned}
[\hat{E}^{rs}_{pq}, \hat{E}^{a\beta}_{ij}] = {} & + \delta^a_p\, \hat{E}^{rs\beta}_{iqj} + \delta^\beta_p\, \hat{E}^{rsa}_{jqi} + \delta^a_q\, \hat{E}^{sr\beta}_{ipj} + \delta^\beta_q\, \hat{E}^{sra}_{jpi} \\
& + \delta^a_p \delta^\beta_q\, \hat{E}^{rs}_{ij} + \delta^\beta_p \delta^a_q\, \hat{E}^{rs}_{ji} \\
& - \delta^r_i\, \hat{E}^{a\beta s}_{pvq} - \delta^s_i\, \hat{E}^{a\beta r}_{qvp} - \delta^r_j\, \hat{E}^{\beta as}_{piq} - \delta^s_j\, \hat{E}^{\beta ar}_{qip} \\
& - \delta^r_i \delta^s_j\, \hat{E}^{a\beta}_{pq} - \delta^s_i \delta^r_j\, \hat{E}^{a\beta}_{qp},
\end{aligned}} \tag{10}$$

so that $[\hat{V},\hat{T}_2]$ is $\frac{1}{4} g^{pq}_{rs} t^{a\beta}_{ij}$ times this bracket. The twelve terms fall into three groups — four with a single delta on the upper pair, two with two deltas and no spectating index, four with a single delta on the lower pair, and two with two deltas — and every one of them is a different operator. As in §2 nothing is cancelled: the contracted form is obtained by letting each delta eat its index of g, term by term, which is what the collection needs.

Contracting each delta with g — one index of g per delta, exactly as in the $(2,1)$ block — turns (10) into

$$\begin{aligned}
[\hat{V},\hat{T}_2] = \tfrac{1}{4}\, t^{a\beta}_{ij} \Big({} & + g^{aq}_{rs}\, \hat{E}^{rs\beta}_{iqj} + g^{\beta q}_{rs}\, \hat{E}^{rsa}_{jqi} + g^{pa}_{rs}\, \hat{E}^{sr\beta}_{ipj} + g^{p\beta}_{rs}\, \hat{E}^{sra}_{jpi} \\
& + g^{a\beta}_{rs}\, \hat{E}^{rs}_{ij} + g^{\beta a}_{rs}\, \hat{E}^{rs}_{ji} \\
& - g^{pq}_{is}\, \hat{E}^{a\beta s}_{pvq} - g^{pq}_{ri}\, \hat{E}^{a\beta r}_{qvp} - g^{pq}_{js}\, \hat{E}^{\beta as}_{piq} - g^{pq}_{rj}\, \hat{E}^{\beta ar}_{qip} \\
& - g^{pq}_{ij}\, \hat{E}^{a\beta}_{pq} - g^{pq}_{ji}\, \hat{E}^{a\beta}_{qp} \Big),
\end{aligned} \tag{11}$$

twelve terms, and the last four of them — the ones that carry two deltas — must not be dropped. They are not small and they do not cancel against anything: checked on matrices, the eight single-delta terms of (11) reproduce $[\hat{V},\hat{T}_2]$ only to a relative 4.1×10^{-1}, the four double-delta terms only to 5.9×10^{-1}, and the twelve together to 3.8×10^{-16}. The same holds for the $(2,2)$ rule itself, whose four double-delta terms are as necessary as its eight single-delta ones.

4. What the two blocks contribute

block	rule	terms	rank out
$[\hat{F},\hat{T}_1]$	$(1,1)$	2	1
$[\hat{F},\hat{T}_2]$	$(1,2)$	4	2
$[\hat{V},\hat{T}_1]$, (6)	$(2,1)$	4	2
$[\hat{V},\hat{T}_2]$, (10)	$(2,2)$	12	2 and 3

Together with the conjugate halves these are the first-order part $[\hat{H}, \hat{A}] = [\hat{H}, \hat{T}] + [\hat{H}, \hat{T}]^\dagger$ of the CT expansion. The $(2, 2)$ block is the only one that produces three-body operators, which is where the $[\ldots]_{1,2}$ truncation of `ct_structure.tex` enters.

5. The check

`tests/test_V_A_commutator.py` builds $\hat{V}, \hat{T}_1, \hat{T}_2$ as explicit spin-free matrices on four orbitals ($\{0, 1\}$ occupied, $\{2, 3\}$ virtual) with random tensors, forms each block from the rules term by term, and compares with the matrix commutators:

block	terms per assignment	max$\lvert\Delta\rvert$	relative
$[\hat{V}, \hat{T}_1]$	4	8.9×10^{-15}	6.1×10^{-16}
$[\hat{V}, \hat{T}_2]$	12	5.0×10^{-14}	8.9×10^{-16}

Each block is checked with the terms kept separate, so the listing in the note is verified as it stands rather than through an equivalent folded form.

6. Why the terms stay separate

The same point as in the Fock case, and it is worth stating once here because the $(2, 2)$ block makes it sharper. The twelve terms of (10) are different operators — $\hat{E}^{rs\beta}_{iqj}$ and $\hat{E}^{rsa}_{jqi}$, for instance, agree neither in their upper nor in their lower index pattern — so each restricts the general range to a different block and carries its own coefficient. Some of them have the same *sum* after a relabelling of the dummy labels; that relabelling relates the sums, not the terms, and it is the terms that the collection reads off. Cancelling a pair would merge two coefficients belonging to two different index patterns, and the resulting $\bar{h}$ and $\bar{g}$ would be missing blocks.

References

[1] W. Kutzelnigg and D. Mukherjee, *Normal order and extended Wick theorem for a multiconfiguration reference wave function*, J. Chem. Phys. **107**, 432 (1997).

[2] T. Yanai and T. Shiozaki, *Canonical transcorrelated theory with projected Slater-type geminals*, J. Chem. Phys. **136**, 084107 (2012).